TABLEAU PORTFOLIO PROJECT

GROUP E: BENFORD'S LAW FRAUD DETECTION & VALIDATION

Benford's Law Fraud Detection & Validation

Statistical fraud analysis: Madoff financial records vs. expected Benford distribution

Tableau | Benford's Law | Fraud Detection | Bar+Line Combo | Dual Axis | Financial Analytics | Story

Project Summary

A 4-view Tableau workbook applying Benford's Law to fraud detection, culminating in an analysis of Bernie Madoff's actual financial records vs. the theoretically expected leading-digit distribution. Demonstrates dual-axis bar+line combination charts, deviation analysis, annotated story views, and a tabbed Tableau Story presentation. Shows how digit-frequency analysis can reveal statistical fingerprints of financial fraud.

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1 PROJECT OVERVIEW

Workbook:	Benford's Law Fraud Validation
Views:	4 (Base Case, Madoff Records, Deviation Chart, Story Presentation)
Data Sources:	Natural orders dataset (base case) + Bernie Madoff financial records (fraud case)
Statistical Method:	Benford's Law -- leading digit frequency analysis
Chart Types:	Dual-axis bar + line combo, Deviation bar chart (above/below zero), Tableau Story
Key Finding:	Madoff's digit '1' frequency: ~39% actual vs. 30.10% expected -- 9.29% excess
Business Use Case:	Financial audit, fraud detection, regulatory compliance analytics

2 WHAT IS BENFORD'S LAW?

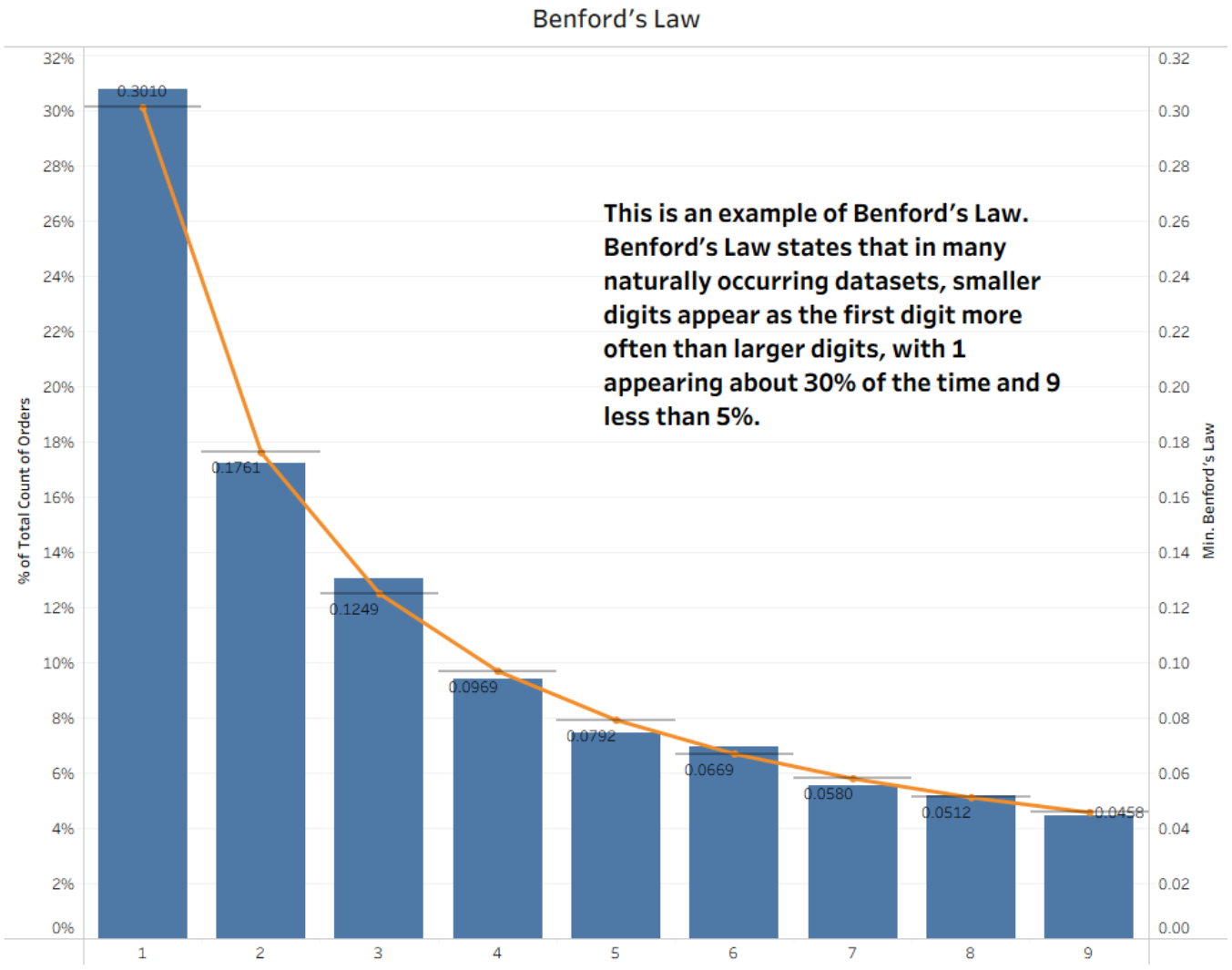
Benford's Law (also called the First-Digit Law) states that in many naturally occurring datasets -- financial transactions, population figures, earthquake magnitudes, river lengths -- the leading digit follows a logarithmic distribution, NOT a uniform one. Smaller digits appear first far more often than larger digits.

Leading Digit	Expected % (Benford)	Formula
1	30.10%	log10(1 + 1/1)
2	17.61%	log10(1 + 1/2)
3	12.49%	log10(1 + 1/3)
4	9.69%	log10(1 + 1/4)
5	7.92%	log10(1 + 1/5)
6	6.69%	log10(1 + 1/6)
7	5.80%	log10(1 + 1/7)
8	5.12%	log10(1 + 1/8)
9	4.58%	log10(1 + 1/9)

Fraudsters who fabricate numbers often unconsciously over-distribute digits, generating too many '1's or creating patterns inconsistent with Benford's expected curve. This makes Benford's Law a powerful -- and surprisingly simple -- first-pass fraud detector.

3 VIEW 1 -- BENFORD'S LAW: BASE CASE (NATURAL DATA)

Bar chart showing % of Total Count of Orders by leading digit (1-9), with an orange line overlay showing the theoretical Benford curve. Natural order data closely follows the expected distribution -- bars align tightly with the orange curve, confirming the dataset is unmanipulated.

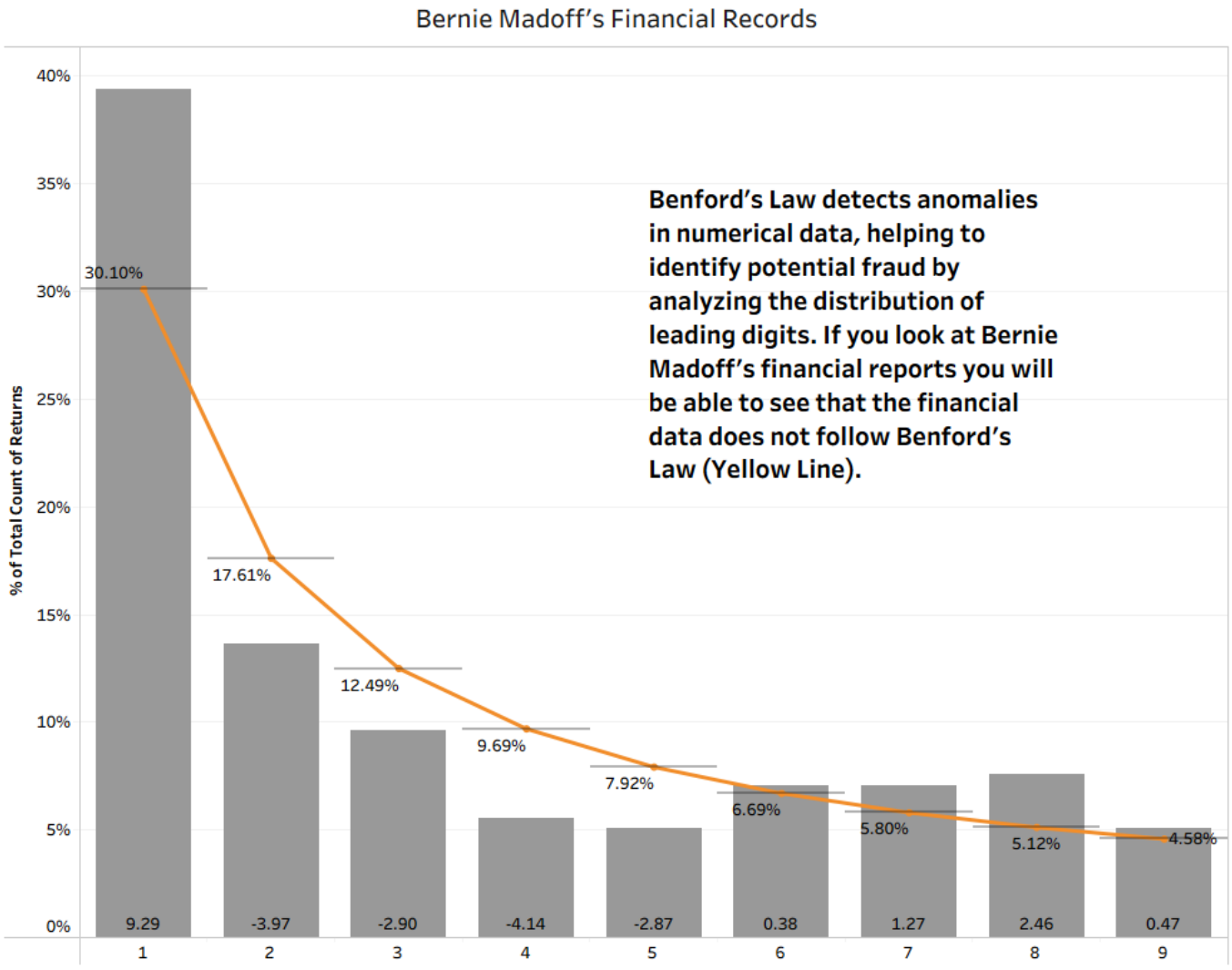


Benford's Law -- Natural orders dataset. Blue bars = actual leading digit frequency. Orange line = theoretical Benford distribution. Digit 1: 30.10% actual vs 30.10% expected -- near-perfect fit.

Tableau technique: Dual-axis chart with SUM([Count]) as bars on primary axis and MIN([Benford Expected]) as a line on the secondary axis. Axes synchronized. Annotation box added via Annotate > Area to explain Benford's Law to dashboard viewers.

4 VIEW 2 -- BERNIE MADOFF'S FINANCIAL RECORDS

The same dual-axis bar+line chart applied to Madoff's actual reported financial figures. Immediately visible: digit '1' appears ~39% of the time (vs. 30.10% expected), a +9.29% excess. Digits 2-5 appear less often than Benford predicts (all below the orange line). This statistical anomaly is a hallmark pattern of fabricated financial data.

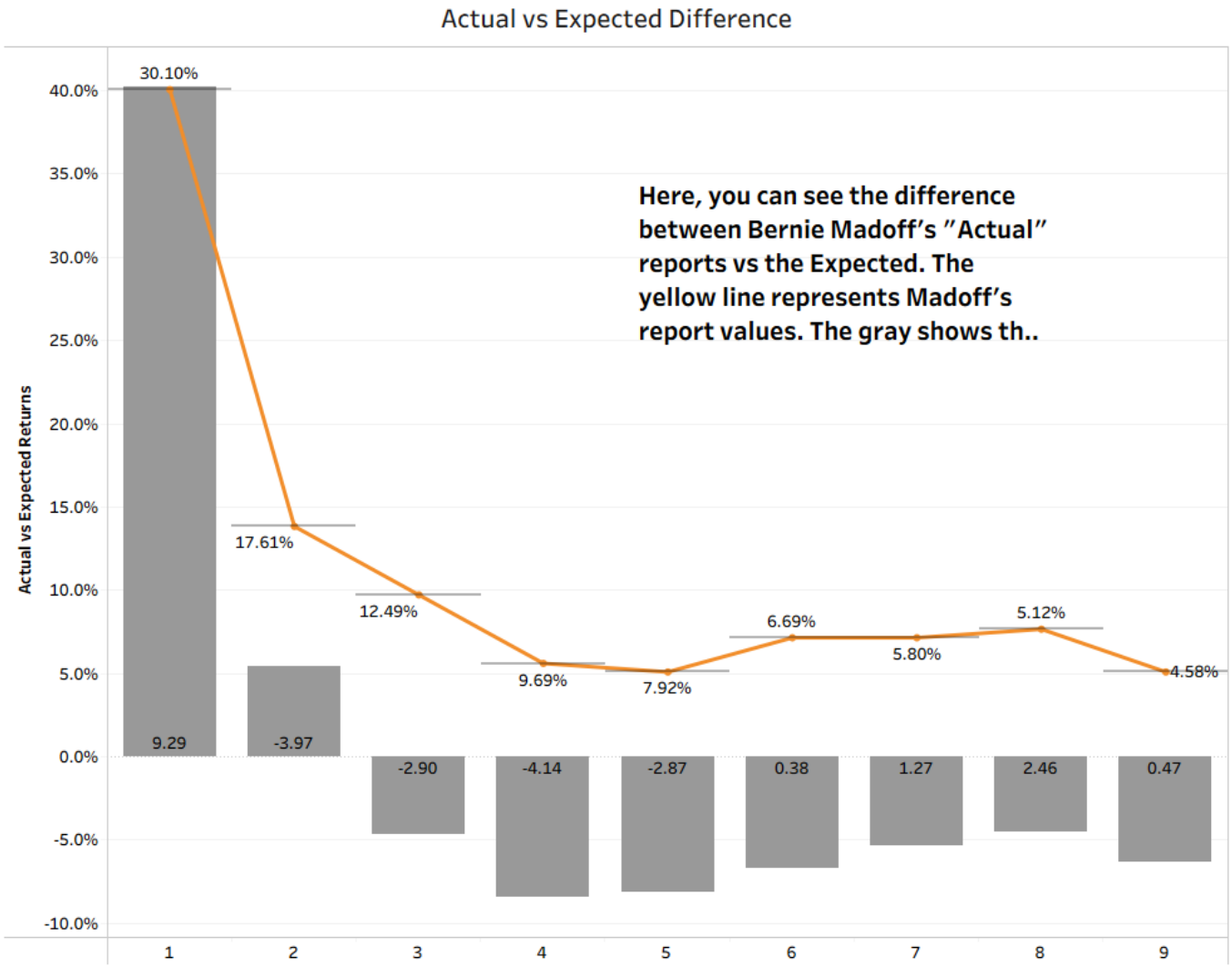


Bernie Madoff's Financial Records -- Actual leading digit distribution vs. Benford expected curve (orange). Digit 1: ~39% actual vs. 30.10% expected. Digits 3-5 below expected. Clear deviation from natural distribution.

- Digit 1: +9.29% excess -- fraudsters commonly start fabricated numbers with 1 (e.g., $1,000, $1.2M).
- Digit 2: -3.97% deficit -- underrepresentation of 2-starting values.
- Digits 3-5: All below expected -- suggest systematic avoidance or clustering around 1-prefix values.
- Digit 8: +2.46% slight excess -- may indicate fabrication of large round numbers starting with 8.
- Overall pattern: Bars diverge from the orange Benford line at multiple digits -- statistically improbable in a legitimate dataset of this size.

5 VIEW 3 -- ACTUAL vs EXPECTED DIFFERENCE: DEVIATION ANALYSIS

A deviation bar chart showing Madoff's actual digit frequency MINUS Benford's expected frequency for each digit. Bars above zero indicate digits that appear more often than expected; bars below zero indicate underrepresentation. The orange line shows Madoff's absolute values for context. Digit 1 stands out at +9.29 percentage points above Benford -- the largest single deviation.

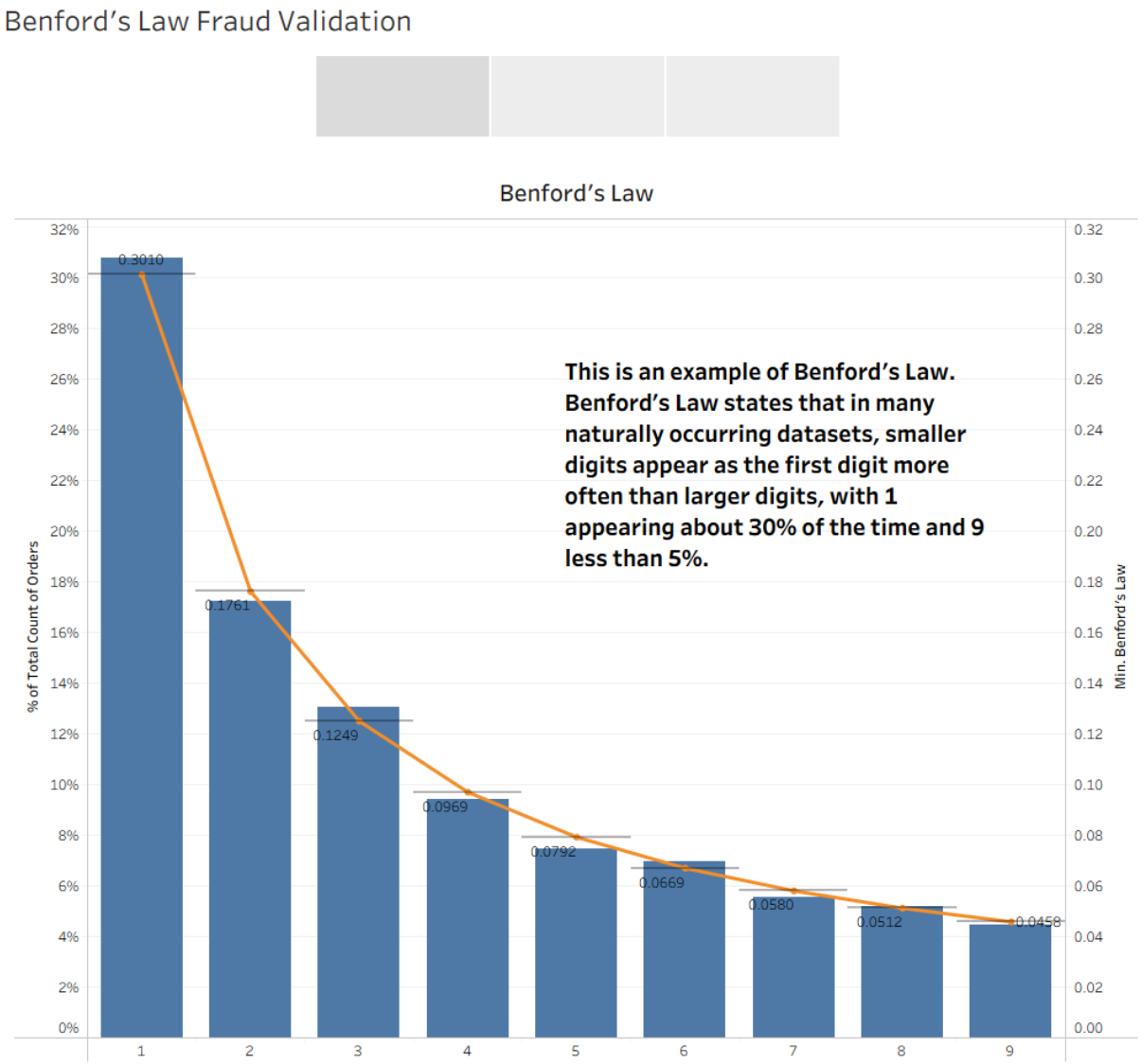


Actual vs Expected Difference -- Deviation of Madoff's digit frequencies from Benford's expected values. Digit 1: +9.29 (above). Digits 3,4,5: negative (below). Orange line = Madoff actual values.

Tableau technique: Calculated field [Deviation] = [Madoff Actual %] - [Benford Expected %]. Bar color uses a diverging color calculation: IF [Deviation] >= 0 THEN 'Above' ELSE 'Below' END -- encoded with distinct colors for immediate visual parsing.

6 VIEW 4 -- TABLEAU STORY: FRAUD VALIDATION PRESENTATION

A Tableau Story assembles the full Benford's Law fraud analysis into a 3-tab navigable presentation: Benford's Law (base case) | Bernie Madoff's Financial Records | Actual vs Expected Difference. Each tab walks viewers through the statistical argument sequentially, from establishing the baseline to revealing the fraud signature.



Benford's Law Fraud Validation -- Tableau Story with tab navigation. Shown: Benford's Law base case tab. Tabs: Benford's Law | Bernie Madoff's Financial Records | Actual vs Expected Difference.

7 SKILLS DEMONSTRATED

- Benford's Law implementation: LOG10(1 + 1/[Digit]) as a Tableau calculated field for expected distribution.
- Dual-axis bar + line combo: Primary axis for bar (count frequency), secondary axis for line (expected curve), synchronized scales.
- Deviation chart: Calculated difference field with above/below-zero diverging color encoding.
- Annotation boxes: Annotate > Area in Tableau for explanatory callout text embedded in the view.
- Tableau Story: 3-step narrative with tab navigation presenting a complete fraud detection case study.
- Financial analytics application: Connecting statistical theory (Benford's Law) to a real-world fraud case (Madoff, $65B Ponzi scheme).
- Dual color encoding: Matching Madoff bars to grey (representing fabricated data) vs. blue (natural data) for visual contrast

across views.